

Lecture 2

Structure of groups, and a first look at crystallographic groups

Objectives. We learn to compare groups by structure-preserving maps and to take groups apart: homomorphisms and their kernels, conjugacy and its geometric meaning, quotient groups and the first isomorphism theorem, direct products, and the construction of semidirect products from a normal subgroup and an action, and the decomposition of a finite group into simple constituents. As a first physical application we prove the crystallographic restriction theorem: a crystal lattice admits only two-, three-, four- and sixfold rotation axes.

2.1 Homomorphisms

Lecture 1 declared two groups “the same” when a bijection matches their multiplication tables. Far more common, and at least as useful, are maps that preserve multiplication *without* being bijective: they compress a group onto a smaller one, retaining some structure and discarding the rest.

Definition 2.1 (Homomorphism). A map $\varphi : G \rightarrow H$ between groups is a *homomorphism* if

$$\varphi(g_1 g_2) = \varphi(g_1) \varphi(g_2) \quad \text{for all } g_1, g_2 \in G.$$

An *isomorphism* is precisely a bijective homomorphism. The *kernel* and *image* of φ are

$$\ker \varphi = \{ g \in G : \varphi(g) = e_H \}, \quad \text{im } \varphi = \{ \varphi(g) : g \in G \}.$$

Example 2.2. (i) The sign $\text{sgn} : S_n \rightarrow \{\pm 1\}$ is a homomorphism — this is exactly the multiplicativity proved in Lecture 1 — with kernel the alternating group A_n . (ii) The determinant $\det : \text{GL}(n, \mathbb{F}) \rightarrow \mathbb{F} \setminus \{0\}$ is a homomorphism, since $\det(AB) = \det A \det B$; its kernel is the *special linear group* $\text{SL}(n, \mathbb{F})$ of matrices of determinant 1. (iii) Realise D_n as orthogonal transformations of the plane, as in Lecture 1. The restriction of the determinant,

$$\nu : D_n \rightarrow \{\pm 1\}, \quad \nu(g) = \det g = \begin{cases} +1, & g \text{ a rotation,} \\ -1, & g \text{ a reflection,} \end{cases}$$

records the *orientation behaviour* of each symmetry operation. This *orientation homomorphism* is our running example; its kernel is the rotation subgroup $\langle r \rangle \cong C_n$. (iv) Reduction modulo n , the map $\mathbb{Z} \rightarrow \mathbb{Z}_n$ sending an integer to its residue, is a homomorphism of additive groups with kernel $n\mathbb{Z} = \{nk : k \in \mathbb{Z}\}$. (v) The map $\varphi(x) = e^{2\pi i x}$ is a homomorphism from $(\mathbb{R}, +)$ onto the circle group $\text{U}(1) = \{z \in \mathbb{C} : |z| = 1\}$, since $e^{2\pi i(x+y)} = e^{2\pi i x} e^{2\pi i y}$; its kernel is $\mathbb{Z}$. (vi) For any pair G, H the *trivial* homomorphism sends all of G to e_H : kernel G , image $\{e_H\}$.

Proposition 2.3 (Basic properties). Let $\varphi: G \rightarrow H$ be a homomorphism. Then:

- (i) $\varphi(e_G) = e_H$, and $\varphi(g^{-1}) = \varphi(g)^{-1}$; more generally $\varphi(g^k) = \varphi(g)^k$ for all $k \in \mathbb{Z}$. If g has finite order, then $\text{ord}(\varphi(g))$ divides $\text{ord}(g)$;
- (ii) $\text{im } \varphi$ is a subgroup of H , and $\ker \varphi$ is a subgroup of G ;
- (iii) φ is injective if and only if $\ker \varphi = \{e_G\}$;
- (iv) composites of homomorphisms are homomorphisms.

Proof. (i) Since $\varphi(e)\varphi(e) = \varphi(ee) = \varphi(e)$, cancellation in H gives $\varphi(e) = e_H$; then $\varphi(g)\varphi(g^{-1}) = \varphi(gg^{-1}) = e_H$ identifies $\varphi(g^{-1})$ as the inverse of $\varphi(g)$, and $\varphi(g^k) = \varphi(g)^k$ follows by induction in both directions. If g has finite order and $n = \text{ord}(g)$ then $\varphi(g)^n = \varphi(g^n) = e_H$, so $\text{ord}(\varphi(g)) \mid n$ by the order lemma of Lecture 1 (whose “if and only if” used only that the element has finite order, not that the group is finite).

(ii) $\text{im } \varphi$ contains e_H and is closed under products and (by (i)) inverses; likewise if $\varphi(g) = \varphi(k) = e_H$ then $\varphi(gk) = e_H$ and $\varphi(g^{-1}) = e_H$. The subgroup test of Lecture 1 applies.

(iii) If φ is injective, only e_G can map to e_H . Conversely suppose $\ker \varphi = \{e\}$ and $\varphi(g) = \varphi(g')$; then $\varphi(g^{-1}g') = \varphi(g)^{-1}\varphi(g') = e_H$, so $g^{-1}g' = e$, i.e. $g = g'$.

(iv) $(\psi\varphi)(g_1g_2) = \psi(\varphi(g_1)\varphi(g_2)) = \psi\varphi(g_1)\psi\varphi(g_2)$. ■

A homomorphism thus forgets information in a controlled way, and the kernel measures exactly *how much* is forgotten: trivial kernel, nothing lost; kernel $\langle r \rangle$ in the orientation map, everything lost except whether the operation flips the plane over. Making “ G with $\ker \varphi$ forgotten” into a group in its own right is the business of Section 2.3; but first we need the right notion of how a group acts on itself, which is also the key to the physical meaning of conjugation.

2.2 Conjugacy

Definition 2.4 (Conjugacy). Elements g, g' of a group G are *conjugate* if $g' = hgh^{-1}$ for some $h \in G$. Conjugacy is an equivalence relation ($g = ege^{-1}$; if $g' = hgh^{-1}$ then $g = h^{-1}g'h$; and $k(hgh^{-1})k^{-1} = (kh)g(kh)^{-1}$), so G is partitioned into *conjugacy classes*; the class of g is denoted $[g] = \{hgh^{-1} : h \in G\}$.

Conjugation has a sharp physical interpretation: hgh^{-1} is *the operation g as seen in the frame transformed by h* . Indeed, when G consists of transformations of space, the identity

$$(hgh^{-1})(hx) = h(g(x))$$

says that hgh^{-1} does to the transformed point hx exactly what g does to x : transform back, apply g , transform forward. Conjugate operations are therefore *the same kind of operation*, differing only in where they sit — a rotation through θ about one axis is conjugate to the rotation through θ about any other axis the group can reach. Quantities that physics should not care about the placement of — characters, in Lecture 5 — will be constant on conjugacy classes.

Example 2.5 (Conjugation in D_n). From the relations $r^n = s^2 = e$, $sr^ks^{-1} = r^{-k}$ of Lecture 1:

- (i) $sr^ks^{-1} = r^{-k}$: viewed in a mirror, a rotation runs backwards, so r^k and r^{-k} are conjugate;

- (ii) $r^m (r^j s) r^{-m} = r^{j+m} (s r^{-m}) = r^{j+m} r^m s = r^{j+2m} s$: conjugating a reflection by a rotation gives the reflection whose axis is rotated accordingly — the index shifts by *two* because rotating the axis by $2\pi m/n$ moves the mirror image twice as far;
- (iii) $s (r^j s) s^{-1} = (s r^j) (s s^{-1}) = s r^j = r^{-j} s$, by the same rules: reflections conjugate to reflections.

Example 2.6 (Conjugacy classes of D_5). Take $n = 5$, so $|D_5| = 10$. By Example 2.5(i), $[r] = \{r, r^4\}$ and $[r^2] = \{r^2, r^3\}$, and these classes are no larger: conjugating r^k by a rotation does nothing (rotations commute), and by a reflection inverts it. For the reflections, conjugation by r shifts the index $j \mapsto j + 2$ modulo 5, and since 2 generates the residues modulo 5, repeated shifts reach every index: all five reflections form a single class. Hence

$$D_5 = \{e\} \cup \{r, r^4\} \cup \{r^2, r^3\} \cup \{s, rs, r^2s, r^3s, r^4s\}, \quad 10 = 1 + 2 + 2 + 5.$$

Geometrically the single reflection class is plain: every axis of the regular pentagon passes through one vertex and one edge midpoint — all five axes look alike.

Remark 2.7 (D_4 , and even against odd). For even n the reflections split into *two* classes, since the index shift $j \mapsto j + 2$ now reaches only every other reflection: the axes through opposite vertices and the axes through opposite edge midpoints are genuinely different kinds of mirror. For the square,

$$D_4 = \{e\} \cup \{r^2\} \cup \{r, r^3\} \cup \{s, r^2s\} \cup \{rs, r^3s\} :$$

five conjugacy classes — a number worth remembering when we count the irreducible representations of D_4 in Lecture 5.

In symmetric groups, conjugation is literally relabelling:

Lemma 2.8 (Relabelling). For $\sigma \in S_n$ and any cycle,

$$\sigma(a_1 a_2 \dots a_k) \sigma^{-1} = (\sigma(a_1) \sigma(a_2) \dots \sigma(a_k)).$$

Proof. Apply the left-hand side to $\sigma(a_i)$: it is sent to $\sigma(\tau(a_i)) = \sigma(a_{i+1})$, indices modulo k , where τ denotes the cycle. A point b not of the form $\sigma(a_i)$ has $\sigma^{-1}(b) \notin \{a_1, \dots, a_k\}$, so τ fixes it and b is fixed. This is exactly the action of the right-hand side. ■

Proposition 2.9 (Conjugacy in S_n). Two permutations are conjugate in S_n if and only if they have the same cycle type, i.e. the same multiset of cycle lengths in their disjoint cycle decompositions.

Proof. Writing π as a product of disjoint cycles and applying Lemma 2.8 cycle by cycle, $\sigma\pi\sigma^{-1}$ is the product of the relabelled cycles, which are again disjoint: the cycle type is preserved. Conversely, given π, π' of the same type, write their disjoint cycles beneath one another, matched by length (including fixed points as 1-cycles); the map σ sending each symbol of π to the symbol below it is a permutation, and by the lemma $\sigma\pi\sigma^{-1} = \pi'$. ■

Example 2.10 (Conjugacy classes of S_3). The cycle types in S_3 are: identity; one transposition; one 3-cycle. Hence the classes are $\{e\}$, $\{(12), (13), (23)\}$, $\{(123), (132)\}$, of sizes $1 + 3 + 2 = 6$. Under the isomorphism $D_3 \cong S_3$ of Lecture 1 these match the classes $\{e\}$, $\{\text{three reflections}\}$, $\{r, r^2\}$ — as they must: an isomorphism carries classes to classes.

How large can a conjugacy class be? The answer is another application of the coset machinery of Lecture 1.

Definition 2.11 (Centraliser and centre). The *centraliser* of $g \in G$ is $C_G(g) = \{x \in G : xg = gx\}$, a subgroup of G (closure and inverses are immediate). The *centre* of G is $Z(G) = \{x \in G : xg = gx \text{ for all } g \in G\}$, the set of elements commuting with everything; thus G is abelian precisely when $Z(G) = G$, and $[g] = \{g\}$ precisely when $g \in Z(G)$.

Proposition 2.12 (Class sizes). For g in a finite group G ,

$$|[g]| = [G : C_G(g)].$$

In particular every class size divides $|G|$, and $|G| = \sum_i |[g_i]|$, summed over the distinct classes (the class equation).

Proof. Map left cosets to conjugates by $xC_G(g) \mapsto xgx^{-1}$. This is well defined and injective because

$$xgx^{-1} = ygy^{-1} \iff (y^{-1}x)g(y^{-1}x)^{-1} = g \iff y^{-1}x \in C_G(g),$$

which by Lecture 1 is exactly the condition $xC_G(g) = yC_G(g)$; and it is surjective by the definition of $[g]$. So $|[g]|$ equals the number of cosets, which divides $|G|$ by Lagrange's theorem. The class equation restates that the classes partition G . ■

The class equations $10 = 1 + 2 + 2 + 5$ of D_5 and $6 = 1 + 3 + 2$ of S_3 illustrate both assertions. For D_4 , Remark 2.7 gives $8 = 1 + 1 + 2 + 2 + 2$, and the two singleton classes identify the centre: $Z(D_4) = \{e, r^2\}$, while $Z(D_5) = \{e\}$.

2.3 Normal subgroups and quotient groups

Which subgroups can be “forgotten”, i.e. arise as kernels? Kernels have a special stability property: if $k \in \ker \varphi$, then $\varphi(gkg^{-1}) = \varphi(g)e_H\varphi(g)^{-1} = e_H$, so $g(\ker \varphi)g^{-1} \subseteq \ker \varphi$ for every g . Subgroups with this property get a name.

Definition 2.13 (Normal subgroup). A subgroup $N \leq G$ is *normal*, written $N \trianglelefteq G$, if $gNg^{-1} = N$ for all $g \in G$, where $gNg^{-1} = \{gng^{-1} : n \in N\}$.

Proposition 2.14 (Equivalent formulations). For a subgroup $N \leq G$, the following are equivalent:

- (i) $gNg^{-1} \subseteq N$ for all $g \in G$;
- (ii) $gNg^{-1} = N$ for all $g \in G$;
- (iii) $gN = Ng$ for all $g \in G$ (left and right cosets coincide);
- (iv) N is a union of conjugacy classes of G .

Proof. (i) $\Rightarrow$ (ii): applying (i) to g^{-1} gives $g^{-1}Ng \subseteq N$, i.e. $N \subseteq gNg^{-1}$. (ii) $\Rightarrow$ (iii): multiply $gNg^{-1} = N$ on the right by g . (iii) $\Rightarrow$ (i): $gn = n'g$ for some $n' \in N$, so $gng^{-1} = n' \in N$. (ii) $\Leftrightarrow$ (iv): N is stable under all conjugations iff with each element it contains that element's entire class. ■

Example 2.15. (i) In an abelian group every subgroup is normal ($gng^{-1} = n$). The centre $Z(G)$ is normal in any G for the same reason. (ii) Kernels of homomorphisms are normal, as computed above; thus $A_n \trianglelefteq S_n$ (kernel of sgn), $\text{SL}(n, \mathbb{F}) \trianglelefteq \text{GL}(n, \mathbb{F})$ (kernel of $\det$), and $\langle r \rangle \trianglelefteq D_n$ (kernel of the orientation map ν). (iii) Any subgroup of index 2 is normal: if $H \leq G$ with $[G : H] = 2$, then for $g \notin H$ both the left and the right coset of g must be the complement $G \setminus H$, so $gH = Hg$; for $g \in H$ this is trivial. This re-proves $\langle r \rangle \trianglelefteq D_n$ and $A_n \trianglelefteq S_n$ without mentioning a homomorphism. (iv) Reflections in D_3 generate non-normal

subgroups: $rsr^{-1} = r^2s \notin \langle s \rangle = \{e, s\}$. Equivalently, $\{e, s\}$ is not a union of classes — the class of s is all three reflections (Example 2.10).

Normality is exactly what is needed to multiply cosets.

Theorem 2.16 (Quotient group). *Let $N \trianglelefteq G$, and let G/N denote the set of cosets of N in G (left and right cosets agreeing). Then*

$$(gN)(hN) := (gh)N$$

is well defined — independent of the representatives g, h chosen — and makes G/N a group, the quotient group, of order $[G : N]$. The natural projection $\pi: G \rightarrow G/N$, $\pi(g) = gN$, is a surjective homomorphism with $\ker \pi = N$. Conversely, if the recipe above is well defined for a subgroup N , then N is normal: the normal subgroups of G are precisely the kernels of homomorphisms out of G .

Proof. Well-definedness: let $g' = gn$, $h' = hm$ with $n, m \in N$ be other representatives. Then

$$g'h' = gn hm = gh(h^{-1}nh)m \in (gh)N,$$

since $h^{-1}nh \in N$ by normality; so $(g'h')N = (gh)N$. The group axioms for G/N are inherited at once from G : associativity from associativity, identity $eN = N$, inverse $(gN)^{-1} = g^{-1}N$. The number of elements is the number of cosets, $[G : N]$. The projection satisfies $\pi(gh) = (gh)N = (gN)(hN) = \pi(g)\pi(h)$ and is onto, and $\pi(g) = N$ iff $g \in N$.

For the converse, take any $n \in N$ and $g \in G$, and compute the product of the cosets gN and $g^{-1}N$ with two choices of representative: $(g)(g^{-1}) = e$ gives the coset N , while $(gn)(g^{-1}) = gng^{-1}$ must then also lie in N . Hence $gNg^{-1} \subseteq N$ for all g , which is normality by Proposition 2.14. ■

Example 2.17. (i) $\mathbb{Z}/n\mathbb{Z} \cong \mathbb{Z}_n$: the cosets of $n\mathbb{Z}$ in $\mathbb{Z}$ are the residue classes modulo n , and coset addition is addition of residues. (ii) $D_n/\langle r \rangle$ has two elements: the coset of rotations and the coset of reflections. Its multiplication table says: rotation $\times$ rotation = rotation, rotation $\times$ reflection = reflection, reflection $\times$ reflection = rotation. That is the group C_2 , and it is precisely the arithmetic of orientation. The quotient has *coarse-grained* D_n : it retains only the information the orientation homomorphism ν sees. (iii) Likewise, for $n \geq 2$, $S_n/A_n \cong C_2$ is the arithmetic of parity: even/odd $\times$ even/odd.

2.4 The first isomorphism theorem

The pattern of the last example — quotient by the kernel reproduces the image — is completely general, and is the single most used theorem of elementary group theory.

Theorem 2.18 (First isomorphism theorem). *Let $\varphi: G \rightarrow H$ be a homomorphism with kernel $K = \ker \varphi$. Then*

$$\bar{\varphi}: G/K \longrightarrow \text{im } \varphi, \quad \bar{\varphi}(gK) = \varphi(g),$$

is an isomorphism: $G/\ker \varphi \cong \text{im } \varphi$.

Proof. The formula is well defined and injective for the same reason:

$$gK = g'K \iff g^{-1}g' \in K \iff \varphi(g^{-1}g') = e_H \iff \varphi(g) = \varphi(g').$$

It is a homomorphism, $\bar{\varphi}((gK)(hK)) = \bar{\varphi}(ghK) = \varphi(gh) = \varphi(g)\varphi(h)$, and it is surjective onto $\text{im } \varphi$ by construction. ■

Example 2.19. (i) The orientation map $\nu: D_n \rightarrow \{\pm 1\}$ is surjective with kernel $\langle r \rangle$, so $D_n/\langle r \rangle \cong C_2$ — now a one-line consequence rather than a table inspection. Likewise $S_n/A_n \cong C_2$ from sgn , for $n \geq 2$. (ii) $\det: O(n) \rightarrow \{\pm 1\}$ is surjective with kernel $SO(n)$, so $O(n)/SO(n) \cong C_2$: orthogonal transformations come in two orientation components. (iii) $\varphi(x) = e^{2\pi i x}$ maps $(\mathbb{R}, +)$ onto $U(1)$ with kernel $\mathbb{Z}$, so

$$\mathbb{R}/\mathbb{Z} \cong U(1) :$$

winding the real line onto the circle. Quotients of *continuous* groups will be everyday objects in Part II.

2.5 Direct products

Having learnt to take groups apart, we record the simplest way to put them together.

Definition 2.20 (Direct product). The *direct product* of groups G and H is the set $G \times H = \{(g, h) : g \in G, h \in H\}$ with componentwise multiplication $(g, h)(g', h') = (gg', hh')$. It is a group of order $|G| |H|$ (when both are finite).

Proposition 2.21. In $G \times H$, if g and h have finite order, then the order of (g, h) is the least common multiple of $\text{ord}(g)$ and $\text{ord}(h)$. If either component has infinite order, then (g, h) has infinite order. Consequently $C_m \times C_n \cong C_{mn}$ if and only if $\gcd(m, n) = 1$.

Proof. When both component orders are finite, $(g, h)^k = (g^k, h^k)$ equals (e, e) iff $\text{ord}(g) \mid k$ and $\text{ord}(h) \mid k$ (Lecture 1's order lemma, which needs only finite element orders), iff $\text{lcm}(\text{ord } g, \text{ord } h) \mid k$. If either component has infinite order, no positive power of the pair can be the identity, since that would make the corresponding component have finite order. If $\gcd(m, n) = 1$ and g, h generate C_m, C_n , then (g, h) has order $\text{lcm}(m, n) = mn = |C_m \times C_n|$, so the product is cyclic. If $d = \gcd(m, n) > 1$, every element's order divides $\text{lcm}(m, n) = mn/d < mn$, so no generator exists. ■

Thus $C_6 \cong C_2 \times C_3$, while $C_2 \times C_2 \not\cong C_4$: the latter product is our acquaintance the Klein four-group, $V \cong C_2 \times C_2$ (match the three elements of order 2 — *involutions* — of V with $(g, e), (e, h), (g, h)$).

It is often more useful to recognise a direct product *inside* a given group:

Proposition 2.22 (Internal direct product). Let $N, M \trianglelefteq G$ with $N \cap M = \{e\}$ and $NM := \{nm : n \in N, m \in M\} = G$. Then every element of N commutes with every element of M , and

$$N \times M \longrightarrow G, \quad (n, m) \longmapsto nm$$

is an isomorphism.

Proof. For $n \in N, m \in M$, consider $c = nm n^{-1} m^{-1}$. Reading it as $(nm n^{-1}) m^{-1}$, normality of M puts it in M ; reading it as $n(m n^{-1} m^{-1})$, normality of N puts it in N . So $c \in N \cap M = \{e\}$, i.e. $nm = mn$. The map is then a homomorphism: $(nm)(n'm') = nn'mm'$ by the commuting just proved. It is surjective since $NM = G$, and injective since $nm = e$ forces $n = m^{-1} \in N \cap M = \{e\}$. ■

Example 2.23 (Parity: $O(3) \cong SO(3) \times C_2$). In three dimensions let -1 be the *spatial inversion* (parity operation) $\vec{r} \mapsto -\vec{r}$. It commutes with every matrix, so $M = \{1, -1\} \cong C_2$ is a normal

subgroup of $O(3)$; and $N = SO(3)$ is normal as the kernel of $\det$. Since $\det(-1) = (-1)^3 = -1$, inversion is *not* a rotation: $N \cap M = \{1\}$. Finally every $A \in O(3)$ with $\det A = -1$ factors as $A = (-1)(-A)$ with $-A \in SO(3)$, so $NM = O(3)$. By Proposition 2.22,

$$O(3) \cong SO(3) \times C_2.$$

Every orthogonal transformation of space is uniquely a rotation, possibly followed by inversion — the group-theoretic statement that *parity* is an independent, commuting label on top of rotational symmetry. (This will let us treat the parity of quantum states as a separate quantum number in Part II.) The argument visibly uses that the dimension is odd: in the plane, -1 is the rotation through π , it *does* lie in $SO(2)$, and no such splitting of $O(2)$ exists.

Direct products also describe physical systems with independent symmetries — two non-interacting subsystems, or geometric and internal symmetries acting on different attributes of the same object. Point groups containing the inversion $i = -1$ are products $G \times \{E, i\}$ in exactly the manner of Example 2.23; the point group of benzene, called D_{6h} in Schoenflies notation, is of this kind.

Semidirect products

The internal direct product asked both factors to be normal. Dropping that demand from one of them gives a strictly larger and far more common construction — the one behind every symmetry group of a space with a distinguished origin.

Definition 2.24 (Internal semidirect product). Let $N \trianglelefteq G$ and $H \leq G$ with $N \cap H = \{e\}$ and $NH = G$. Then G is the (*internal*) *semidirect product* of N by H , written $G = N \rtimes H$. Each $g \in G$ factors uniquely as $g = nh$ ($n \in N$, $h \in H$): existence is $NH = G$, and if $nh = n'h'$ then $n'^{-1}n = h'h^{-1} \in N \cap H = \{e\}$. Moreover conjugation by H stabilises N : $hnh^{-1} \in N$ for all $h \in H$, $n \in N$.

Only N is required normal; H need not be. When H is also normal the conjugation action is trivial and the construction collapses back to the direct product of Proposition 2.22.

An *automorphism* of N is an isomorphism from N to itself. The automorphisms form a group under composition, denoted $\text{Aut}(N)$.

An *action of H on N by automorphisms* means exactly a homomorphism $\varphi: H \rightarrow \text{Aut}(N)$; writing φ_h for the image of h , the two laws packed into that word are

$$\varphi_e = \text{id}_N, \quad \varphi_{hh'} = \varphi_h \circ \varphi_{h'}.$$

In words: each h twists N by an automorphism, the identity twists nothing, and twisting by a product is twisting twice. This twist is what the next construction inserts into an otherwise ordinary product of pairs.

Proposition 2.25 (External model). Given groups N, H and a homomorphism $\varphi: H \rightarrow \text{Aut}(N)$, written $h \mapsto \varphi_h$, the set $N \times H$ with

$$(n, h)(n', h') = (n \varphi_h(n'), hh')$$

is a group $N \rtimes_{\varphi} H$, with identity (e, e) and inverse $(n, h)^{-1} = (\varphi_{h^{-1}}(n^{-1}), h^{-1})$. Identifying N with $N \times \{e\}$ and H with $\{e\} \times H$ realises Definition 2.24, the action being $\varphi_h(n) = hnh^{-1}$.

Proof. Associativity is the single computation

$$\begin{aligned} ((n, h)(n', h'))(n'', h'') &= (n \varphi_h(n') \varphi_{hh'}(n''), hh'h'') \\ &= (n, h)((n', h')(n'', h'')), \end{aligned}$$

where $\varphi_{hh'} = \varphi_h \circ \varphi_{h'}$ (homomorphism) and each $\varphi_h \in \text{Aut}(N)$ distributes over products. The identity and inverse are immediate. ■

Example 2.26 (The dihedral group). $D_n = \langle r, s \mid r^n = s^2 = e, srs^{-1} = r^{-1} \rangle$ has the rotation subgroup $\langle r \rangle \cong C_n$ normal of index two and the reflection $\langle s \rangle \cong C_2$ meeting it trivially, so $D_n \cong C_n \rtimes C_2$ with C_2 acting on C_n by inversion, $\varphi_s(r) = r^{-1}$.

It is worth running the ordered-pair rule once. Writing elements as (r^a, s^ϵ) , the product of Proposition 2.25 gives

$$(r^a, s^\epsilon)(r^b, s^\delta) = (r^a \varphi_{s^\epsilon}(r^b), s^{\epsilon+\delta}) = (r^{a+(-1)^\epsilon b}, s^{\epsilon+\delta}),$$

which is exactly the multiplication rule for D_n from Lecture 1. The smallest instance shows the twist on its own: $(e, s)(r, e) = (\varphi_s(r), s) = (r^{-1}, s)$ — passing r across s inverts it. The action is nontrivial for $n \geq 3$: dihedral groups are genuinely semidirect, not direct.

The construction carries over unchanged from finite groups to the continuous groups of Part II. The Euclidean group of the plane is $E(2) = \mathbb{R}^2 \rtimes O(2)$ (the translations are normal, acted on by rotations and reflections), and the symmetry group of relativistic spacetime is the Poincaré group $\mathbb{R}^4 \rtimes SO^+(1, 3)$ of Lecture 15 — translations normal, Lorentz transformations acting. In each case the factorisation “a translation followed by a point symmetry” is precisely the unique $g = nh$ of Definition 2.24.

2.6 Simple groups and composition series

Quotients suggest a notion of *atom*: a group that cannot be coarse-grained at all.

Definition 2.27 (Simple group). A group G with $|G| > 1$ is *simple* if its only normal subgroups are $\{e\}$ and G — equivalently, every nontrivial homomorphism out of G is injective.

Proposition 2.28. *An abelian group is simple if and only if it is cyclic of prime order.*

Proof. C_p has no subgroups at all besides $\{e\}$ and itself, by Lagrange’s theorem. Conversely, in an abelian group every subgroup is normal, so a simple abelian G may have no proper nontrivial subgroup whatsoever. Pick $g \neq e$: then $\langle g \rangle = G$, and g cannot have infinite order (else $\langle g^2 \rangle$ would be proper and nontrivial), so G is cyclic of finite order n ; and if $n = ab$ with $1 < a < n$ then $\langle g^a \rangle$ is proper and nontrivial. Hence n is prime. ■

Nonabelian simple groups exist and are plentiful: the alternating groups A_n are simple for all $n \geq 5$ (we shall not prove this). The smallest, A_5 of order 60 — the rotation group of the icosahedron — is the smallest nonabelian simple group of all, and its simplicity is, through Galois’s theory, precisely the reason no formula in radicals solves the general quintic equation.

Definition 2.29 (Composition series). A *composition series* of a finite group G is a chain

$$G = G_0 \supseteq G_1 \supseteq \cdots \supseteq G_k = \{e\},$$

in which each G_{i+1} is normal in G_i (not necessarily in G) and each composition factor G_i/G_{i+1} is simple. Every finite group has one: refine any chain until no further proper normal subgroups can be inserted. The refinement stops because G is finite, and when it stops every factor G_i/G_{i+1} is simple. For suppose Q were a proper nontrivial normal subgroup of G_i/G_{i+1} , and let $\pi: G_i \rightarrow G_i/G_{i+1}$ be the projection. Then $\pi^{-1}(Q)$ is normal in G_i : for $g \in G_i$ and $x \in \pi^{-1}(Q)$, $\pi(gxg^{-1}) = \pi(g)\pi(x)\pi(g)^{-1} \in Q$ because Q is normal. And $G_{i+1} \subsetneq \pi^{-1}(Q) \subsetneq G_i$, since Q is nontrivial and proper. So the chain could be refined further, contrary to maximality.

Example 2.30 (Composition series of D_4). Consider

$$D_4 \supseteq \langle r \rangle \supseteq \langle r^2 \rangle \supseteq \{e\}.$$

Each step has index 2, hence is normal in the preceding group by Example 2.15(iii), and each factor is a group of order 2:

$$D_4/\langle r \rangle \cong C_2, \quad \langle r \rangle/\langle r^2 \rangle \cong C_2, \quad \langle r^2 \rangle/\{e\} \cong C_2,$$

all simple. A quite different-looking series uses the Klein subgroup $V_1 = \{e, r^2, s, r^2s\}$ of Sheet 1:

$$D_4 \supseteq V_1 \supseteq \langle s \rangle \supseteq \{e\},$$

again with all indices 2 ($\langle s \rangle$ is normal in the *abelian* group V_1 , though not in D_4 !). Both series have composition factors C_2, C_2, C_2 .

Example 2.31 (C_{12}). For $C_{12} = \langle g \rangle$, the chains $C_{12} \supseteq \langle g^2 \rangle \supseteq \langle g^4 \rangle \supseteq \{e\}$ and $C_{12} \supseteq \langle g^3 \rangle \supseteq \langle g^6 \rangle \supseteq \{e\}$ are composition series with factors C_2, C_2, C_3 and C_3, C_2, C_2 respectively: the factors arrive in a different *order*, but as a multiset they agree — the “prime factorisation” $12 = 2 \cdot 2 \cdot 3$ in group-theoretic form.

This stability is general:

Theorem 2.32 (Jordan–Hölder; statement only). *Any two composition series of a finite group have the same length and the same composition factors with the same multiplicities, up to reordering and isomorphism.*

Finite groups are thus built from simple groups as molecules from atoms, and the classification of finite simple groups mentioned in Lecture 1 is the periodic table. But the analogy has a sting: knowing the atoms does not determine the molecule. C_4 and the Klein four-group both have composition factors C_2, C_2 , yet are not isomorphic — assembling a group from given factors (the *extension problem*) has in general many inequivalent solutions, and no complete classification of all finite groups exists or is expected.

2.7 A first look at crystallographic groups

We now put the structure theory to work on a physical question: *which rotational symmetries can a crystal have?*

An idealised crystal is an infinite periodic array of atoms. Its periodicity is encoded by a *lattice*: fix linearly independent vectors $\vec{a}_1, \vec{a}_2, \vec{a}_3 \in \mathbb{R}^3$ (the *basis* or *primitive vectors*) and set

$$\Lambda = \{ n_1 \vec{a}_1 + n_2 \vec{a}_2 + n_3 \vec{a}_3 : n_1, n_2, n_3 \in \mathbb{Z} \},$$

the set of integer combinations (Figure 2.1 shows the two-dimensional analogue). Translating the crystal through any $\vec{t} \in \Lambda$ maps it onto itself. Besides translations, the crystal may

admit *point* symmetries — rotations and reflections about a fixed origin — and these must at the very least map the lattice onto itself. We therefore study

$$P(\Lambda) = \{ R \in O(3) : R\Lambda = \Lambda \},$$

the *point group of the lattice* (a subgroup of $O(3)$: products and inverses of lattice-preserving maps preserve the lattice).

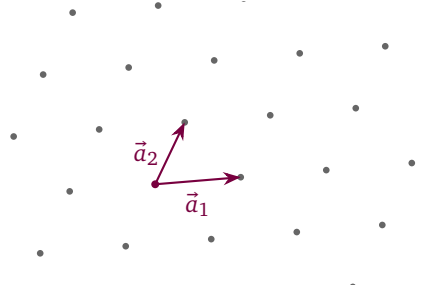


Figure 2.1: A (two-dimensional) lattice: all integer combinations of the primitive vectors. The lattice is infinite and discrete, and the choice of primitive vectors generating it is not unique.

From a crystal to its lattice

The passage from a crystal to the lattice Λ deserves spelling out, because the lattice is a bookkeeping device and its points need not be atoms.

A crystal is specified by two pieces of data: a lattice Λ , and a finite list of atoms with their species and their positions inside one cell — the *motif*. (Crystallographers call the motif the *basis* of the crystal, a second use of a word already spent above on the primitive vectors; we say motif to keep the two apart.) The crystal is then every translate of the motif by every lattice vector. Read in this direction the lattice comes first; but a real crystal arrives as a set of atoms, and the lattice has to be recovered from it. It is recovered as the set of translations that leave the crystal alone,

$$\Lambda = \{ \vec{t} \in \mathbb{R}^3 : \vec{t} + C = C \},$$

C denoting the crystal — a set of atoms with their species, which a translation must preserve species by species. This is the honest transcription of “periodic”: Λ is a group by construction, and the primitive vectors are any choice of generators for it.

The atoms themselves need not form a lattice, and the origin may be put where no atom sits. The standard example is two-dimensional.

Example 2.33 (Honeycomb: the atoms are not the lattice). Carbon atoms at the vertices of a honeycomb, with bond length d , are the sheet of graphene (Figure 2.2). Every atom has three nearest neighbours at distance d , so all atoms are alike chemically — yet the set of atoms is *not* a lattice. Take an atom as origin and let $\vec{\delta}$ be the vector to one of its neighbours. Then $\vec{\delta}$ is an atom, but $2\vec{\delta}$ is the centre of a hexagon, where there is no atom at all: the set is not closed under addition.

What is periodic is the pattern, not the atoms one at a time. Writing

$$\vec{a}_1 = \sqrt{3}d(1, 0), \quad \vec{a}_2 = \sqrt{3}d\left(\frac{1}{2}, \frac{\sqrt{3}}{2}\right),$$

— two vectors of equal length $\sqrt{3}d$ at 60° — the lattice Λ they generate is triangular, and $\vec{\delta} = (\vec{a}_1 + \vec{a}_2)/3$ satisfies $3\vec{\delta} = \vec{a}_1 + \vec{a}_2 \in \Lambda$. The atoms are the two cosets Λ and $\Lambda + \vec{\delta}$, so the motif is two carbon atoms, at $\vec{0}$ and $\vec{\delta}$: one lattice point, two atoms. The remaining coset $\Lambda + 2\vec{\delta}$ is the set of hexagon centres, and carries no atom.

Nothing forces the origin onto an atom. We may equally mark the cells from the hexagon centres, the points of $\Lambda + 2\vec{\delta}$, and describe the same crystal with a motif of two atoms, neither of which sits at a marked point. Note that $\Lambda + 2\vec{\delta}$ is not itself a lattice in the sense above — it is not a group, since it does not contain $\vec{0}$ — but a translate of one, an *affine* set of cell origins. Moving the origin changes the coordinates and the motif; it does not change Λ , which is the group of translations carrying the crystal to itself and is the same whichever origin is chosen. This is the general situation: a marked point records where the motif is repeated, not where matter is.

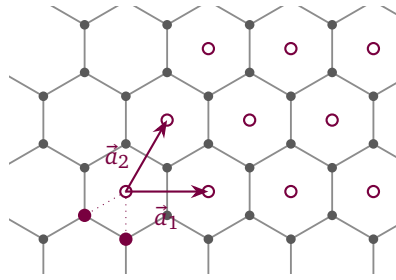


Figure 2.2: The honeycomb. Grey dots are carbon atoms; open circles mark the cell origins, taken here at the hexagon centres, where there is no atom; the vectors between any two of them are exactly the lattice Λ . The two filled atoms are the motif carried by the marked lattice point — one from each sublattice — and translating that pair through Λ produces every atom. The atoms themselves do not form a lattice.

Once the lattice is in hand the symmetry is read off in two stages. The lattice has its own point group $P(\Lambda)$, computed below and depending only on Λ . The crystal's point group about a chosen point is

$$P(C, \vec{x}_0) = \{ R \in O(3) : R(C - \vec{x}_0) = C - \vec{x}_0 \},$$

and it is a subgroup of $P(\Lambda)$: an operation fixing the crystal permutes its translations, hence fixes Λ . It can be a proper subgroup, because the motif need not be as symmetric as the lattice — which is why crystals occur whose point group lacks the inversion that every $P(\Lambda)$ contains (Theorem 2.38).

It also depends on the point chosen. For the honeycomb, $P(\Lambda)$ has order 12 for the triangular lattice; about a hexagon centre the crystal keeps all 12, since the sixfold rotation there exchanges the two atoms of the motif. About an atom it keeps only half of them: of the rotations only the multiples of $2\pi/3$ survive, and of the six mirror lines only the three through that atom, giving a group of order 6 — a copy of D_3 . A sixfold axis and a threefold axis both run through this one crystal, through different points — which is exactly why the full symmetry of a crystal is described not by a point group alone but by a space group, combining point operations with translations.

Lemma 2.34. $P(\Lambda)$ is a finite group.

Proof. First, only finitely many lattice vectors lie in any ball: writing $\vec{v} = B\vec{n}$ with B the

matrix of primitive vectors and $\vec{n} \in \mathbb{Z}^3$, the bound $|\vec{v}| \leq L$ confines $\vec{n} = B^{-1}\vec{v}$ to a bounded region of $\mathbb{R}^3$, which contains finitely many integer points. Now let S be the (finite) set of lattice vectors of length at most $\max_i |\vec{a}_i|$; note S contains the basis vectors, so S spans $\mathbb{R}^3$. Each $R \in P(\Lambda)$ preserves lengths and the lattice, hence permutes S ; and a linear map is determined by its values on a spanning set, so distinct elements of $P(\Lambda)$ induce distinct permutations of S . Thus $P(\Lambda)$ injects into the finite group of permutations of S . ■

So far, so qualitative. The quantitative surprise is how *few* rotations survive. Recall that every rotation $R \in \text{SO}(3)$ is a rotation through some angle θ about some axis (a fact we prove properly in Part II; for now, diagonalise or take it as known), so that in a suitable orthonormal basis adapted to the axis

$$R \sim \begin{pmatrix} \cos \theta & -\sin \theta & 0 \\ \sin \theta & \cos \theta & 0 \\ 0 & 0 & 1 \end{pmatrix}, \quad \text{whence} \quad \text{tr } R = 1 + 2 \cos \theta,$$

the trace being independent of the basis used to compute it.

Theorem 2.35 (Crystallographic restriction). *Let $R \in P(\Lambda)$ be a rotation through the angle θ . Then*

$$\theta \in \left\{0, \frac{\pi}{3}, \frac{\pi}{2}, \frac{2\pi}{3}, \pi\right\} \pmod{2\pi},$$

up to sense of rotation; equivalently, every rotation in $P(\Lambda)$ has order 1, 2, 3, 4 or 6. A lattice admits no fivefold axis, and none of order seven or higher.

Proof. Since $R\Lambda = \Lambda$, each $R\vec{a}_j$ is again a lattice vector: $R\vec{a}_j = \sum_i M_{ij}\vec{a}_i$ with $M_{ij} \in \mathbb{Z}$. In matrix form $RB = BM$, i.e. $M = B^{-1}RB$: in the lattice basis, R is an integer matrix. But the trace is invariant under change of basis, $\text{tr } R = \text{tr}(B^{-1}RB) = \text{tr } M \in \mathbb{Z}$, so

$$1 + 2 \cos \theta \in \mathbb{Z} \implies \cos \theta \in \left\{-1, -\frac{1}{2}, 0, \frac{1}{2}, 1\right\},$$

the only values of cosine in $[-1, 1]$ with $2 \cos \theta$ an integer. These give $\theta = \pi, \frac{2\pi}{3}, \frac{\pi}{2}, \frac{\pi}{3}, 0$: rotations of order 2, 3, 4, 6, 1 respectively. The same argument in the plane, where $\text{tr } R = 2 \cos \theta$, yields the same list. ■

Note how the proof rides on a single structural observation: one operator, two bases. In the orthonormal basis R displays its angle; in the lattice basis it displays its integrality; the trace, blind to basis, forces the two displays to be compatible. This “evaluate an invariant in two ways” manoeuvre will return many times, most spectacularly with characters in Lecture 5.

Remark 2.36 (Fivefold symmetry and quasicrystals). The missing fivefold axis is the group-theoretic face of a familiar fact: regular pentagons do not tile the plane. Nature, however, found a loophole. In 1982 Shechtman observed metallic alloys whose diffraction patterns show sharp peaks with tenfold symmetry — crisp long-range *order* without lattice *periodicity*. These *quasicrystals* (Nobel Prize in Chemistry, 2011), like the Penrose tilings that model them, evade the restriction theorem by giving up the lattice Λ , not by violating the mathematics: the theorem is about periodic structures and says nothing about aperiodic ones.

The classification logic

The restriction theorem is the main step of a complete classification, whose shape we now sketch; the details belong to crystallography, but the logic is pure group theory.

First, one needs the finite rotation groups available in three dimensions — the analogue of Leonardo's theorem from Lecture 1:

Theorem 2.37 (Finite subgroups of $SO(3)$; statement only). *Every finite subgroup of $SO(3)$ is one of: the cyclic groups C_n ; the dihedral groups D_n (realised by n -gons lying in space — in three dimensions the reflection of Lecture 1 is carried out instead by a half-turn about an axis lying in the polygon's plane, which is a proper rotation, so the whole group sits inside $SO(3)$ — and extended to $n = 2$: D_2 is the Klein four-group of half-turns about three perpendicular axes); the rotation group $T \cong A_4$ of the regular tetrahedron (order 12); the rotation group $O \cong S_4$ of the cube and octahedron (order 24); and the rotation group $Y \cong A_5$ of the icosahedron and dodecahedron (order 60).*

Molecules realise the exotic entries: methane's rotations form T , sulphur hexafluoride's form O , and the fullerene C_{60} realises Y . For *crystals*, however, Theorem 2.35 expels every group containing a fivefold rotation or any rotation of order ≥ 7 : the icosahedral group is forbidden, and only $n \in \{1, 2, 3, 4, 6\}$ survive in the cyclic and dihedral families. Adjoining improper operations costs nothing new: every lattice satisfies $-\Lambda = \Lambda$, so the inversion $-\mathbb{1}$ lies in $P(\Lambda)$ for every lattice, and any improper A factors through it as $A = (-\mathbb{1})(-A)$ with $-A$ a rotation, exactly as in Example 2.23 — so the improper possibilities are governed by the proper ones. Assembling the allowed proper and improper combinations yields:

Theorem 2.38 (Hessel; statement only). *Up to equivalence there are exactly 32 finite subgroups of $O(3)$ compatible with a three-dimensional lattice — that is, subgroups of the point group $P(\Lambda)$ of some lattice Λ ; a crystal need not use all of its lattice's symmetry, which is why groups without the inversion occur although every $P(\Lambda)$ contains it: the 32 crystallographic point groups, or crystal classes.*

The corresponding count of essentially different lattices is the 14 *Bravais lattices*; and combining point operations with translations (including the hybrid screw rotations and glide reflections) yields the 230 *space groups* of Fedorov and Schoenflies — the complete catalogue of symmetries available to any crystal in nature. In two dimensions the same programme gives 10 point groups, 5 Bravais lattices and the 17 wallpaper groups. We will not need these lists; what we will need, from Lecture 6 onwards, is the representation theory of the individual point groups, where molecules like H_2O and NH_3 from Lecture 1 reappear with their vibrational spectra. The bridge from groups to spectra — linear representations — is the subject of Lecture 3.

Remark 2.39 (Computer algebra). The structure theory of this lecture is well suited to machine verification. With G a permutation group as in Lecture 1, `GroupCentralizer[G, g]` returns the centraliser of g , whence class sizes via Proposition 2.12; the classes themselves are assembled in a few lines by collecting xgx^{-1} over `GroupElements[G]` with `PermutationProduct` and `InversePermutation`. Normality of a subgroup is checked by conjugating its generators by the generators of G . The accompanying notebook verifies the class equations of D_5 and D_4 and both composition series of Example 2.30. Mathematica's left-to-right composition convention, flagged in Lecture 1, applies here too.

Summary.

- *Homomorphism.* A product-preserving map; its kernel is normal and its image a subgroup (Definition 2.1, Proposition 2.3).
- *Conjugacy.* Conjugacy classes partition the group; in S_n they are the cycle types (Proposition 2.9), and class sizes divide $|G|$ (Proposition 2.12).
- *Quotients.* A normal subgroup yields a quotient group (Theorem 2.16); the first isomorphism theorem gives $G/\ker \varphi \cong \text{im } \varphi$ (Theorem 2.18).
- *Building groups.* Direct products combine independent factors; semidirect products $N \rtimes_{\varphi} H$ encode an action $\varphi: H \rightarrow \text{Aut}(N)$ (Definitions 2.20 and 2.24, Proposition 2.25). Simple groups and the Jordan–Hölder composition series (Theorem 2.32).
- *Crystallographic restriction.* A discrete lattice symmetry admits only 2-, 3-, 4-, and 6-fold axes (Theorem 2.35); the finite subgroups of $\text{SO}(3)$ are classified (Theorem 2.37).

Tutorial. Exercise Sheet 2.

References

Armstrong, *Groups and Symmetry* (cosets, quotient groups, direct and semidirect products);
Jones, *Groups, Representations and Physics*, Pt. 1; Tung, *Group Theory in Physics*, Ch. 1.

Bibliography

The per-lecture *References* sections cite the works below by author and short title; full details are collected here. Chapter and section numbers in the lecture references follow these editions.

- M. A. Armstrong, *Groups and Symmetry*. Springer (UTM), 1988.
- V. Bargmann, On unitary ray representations of continuous groups. *Annals of Mathematics* **59** (1954), 1–46.
- A. O. Barut and R. Rączka, *Theory of Group Representations and Applications*, 2nd ed. World Scientific, 1986.
- R. Berndt, *Representations of Linear Groups: An Introduction Based on Examples from Physics and Number Theory*. Vieweg, 2007. DOI: 10.1007/978-3-8348-9401-4. Theorem 5.2, p. 71 gives the SNAG theorem used for the translation subgroup.
- D. M. Bishop, *Group Theory and Chemistry*. Dover, 1993 (orig. Oxford University Press, 1973).
- O. Bratteli and D. W. Robinson, *Operator Algebras and Quantum Statistical Mechanics 2: Equilibrium States, Models in Quantum Statistical Mechanics*, 2nd ed. (1997), 2nd printing. Springer, 2002. Section 5.2.1, pp. 7–8 gives Fock-space second quantisation, its finite-particle analytic core and closure, and the lifted unitary group.
- G. Costa and G. Fogli, *Symmetries and Group Theory in Particle Physics*. Springer (Lecture Notes in Physics 823), 2012.
- J. B. Conway, *A Course in Functional Analysis*, 1st ed. Springer (GTM 96), 1985. Chapter IX, Theorems 6.4 and 6.6 give the strong/weak bicommutant closure and the multiplication maximal-abelian theorem for sigma-finite measure spaces.
- J. P. Elliott and P. G. Dawber, *Symmetry in Physics*, 2 vols. Macmillan, 1979.
- K. Erdmann and M. J. Wildon, *Introduction to Lie Algebras*. Springer (SUMS), 2006.
- W. Fulton and J. Harris, *Representation Theory: A First Course*. Springer (GTM 129), 1991.
- G. B. Folland, *Harmonic Analysis in Phase Space*. Princeton University Press (Annals of Mathematics Studies 122), 1989. Chapter 1, Sections 2–5, including Theorem 1.50, treats the Heisenberg group, Schrödinger/Weyl representation, and Stone–von Neumann theorem.
- G. B. Folland, *A Course in Abstract Harmonic Analysis*, 2nd ed. Chapman & Hall/CRC, 2016. Section 6.6, Theorem 6.42, pp. 199–201 gives the regular semidirect-product Mackey theorem, including induced irreducibility, exhaustivity, and equivalence.
- R. Gilmore, *Lie Groups, Lie Algebras, and Some of Their Applications*. Wiley, 1974 (repr. Dover, 2005).
- B. C. Hall, *Lie Groups, Lie Algebras, and Representations: An Elementary Introduction*, 2nd ed. Springer (GTM 222), 2015.
- N. Jeevanjee, *An Introduction to Tensors and Group Theory for Physicists*, 2nd ed. Birkhäuser, 2015.
- H. F. Jones, *Groups, Representations and Physics*, 2nd ed. Institute of Physics Publishing,

1998. Chapter 8, Sections 8.3–8.4, pp. 149–164 is the course’s explicit $SU(3)$ further-reading route; Chapter 10, Section 10.4, pp. 208–216 treats the Poincaré little groups; Appendix E, pp. 279–280 derives the covariant on-shell measure from $\theta(p^0)\delta(p^2 - m^2)d^4p$.
- Z.-Q. Ma, *Group Theory for Physicists*. World Scientific, 2007.
 - A. M. L. Messiah and O. W. Greenberg, Symmetrization Postulate and Its Experimental Foundation. *Physical Review* **136** (1964), B248–B267. DOI: <https://doi.org/10.1103/PhysRev.136.B248>.
 - H. Osborn, *Group Theory*. University of Cambridge, Mathematical Tripos Part III lecture notes (available from the DAMTP lecture-notes pages).
 - M. Reed and B. Simon, *Methods of Modern Mathematical Physics, Vol. I: Functional Analysis*, rev. ed. Academic Press, 1980. Chapter VIII, Theorems VIII.5, VIII.8, VIII.10, and VIII.13 and Section VIII.5, Example 2 cover spectral calculus, Stone’s theorem, invariant-domain generators, strong commutation, and the bounded-CCR obstruction.
 - P. Ramond, *Group Theory: A Physicist’s Survey*. Cambridge University Press, 2010. Chapter 11, Section 11.1.1, pp. 224–231 treats the Pauli–Lubanski Casimirs and the massless little group; pp. 230–231 distinguish finite helicity from continuous spin and record the two integer/half-odd-integer cover sectors.
 - H. Rauch, A. Zeilinger, G. Badurek, A. Wilfing, W. Bauspiess, and U. Bonse, Verification of Coherent Spinor Rotation of Fermions. *Physics Letters A* **54** (1975), 425–427. DOI: [10.1016/0375-9601\(75\)90798-7](https://doi.org/10.1016/0375-9601(75)90798-7).
 - J. Schwichtenberg, *Physics from Symmetry*, 2nd ed. Springer, 2018.
 - J.-P. Serre, *Linear Representations of Finite Groups*. Springer (GTM 42), 1977.
 - J. Stillwell, *Naive Lie Theory*. Springer (UTM), 2008.
 - E. M. Stein and R. Shakarchi, *Fourier Analysis: An Introduction*. Princeton University Press (Princeton Lectures in Analysis I), 2003. Chapter 3, Section 1, Theorem 1.1, pp. 69–79 gives the mean-square completeness result used in Lecture 12.
 - M. Tinkham, *Group Theory and Quantum Mechanics*. McGraw–Hill, 1964 (repr. Dover, 2003).
 - W.-K. Tung, *Group Theory in Physics*. World Scientific, 1985. Chapter 10, Section 10.4, pp. 191–200 treats induced Poincaré representations, Pauli–Lubanski brackets, and massive and massless little groups.
 - S. Weinberg, *The Quantum Theory of Fields, Vol. I: Foundations*. Cambridge University Press, 1995. Section 5.6 treats causal fields and the relativistic spin–statistics connection, a theorem not derived from a rotation phase in this course.
 - S. A. Werner, R. Colella, A. W. Overhauser, and C. F. Eagen, Observation of the Phase Shift of a Neutron Due to Precession in a Magnetic Field. *Physical Review Letters* **35** (1975), 1053–1055. DOI: [10.1103/PhysRevLett.35.1053](https://doi.org/10.1103/PhysRevLett.35.1053).
 - P. Woit, *Quantum Theory, Groups and Representations: An Introduction*. Springer, 2017.